Stanford University
Political Science 350D
Problem Set 3
Due April 20, 2015

1. Consider the hierarchical model where

$$\theta_i|\mu,\tau^2 \sim \text{i.i.d. Normal}(\mu,\tau^2) \qquad y_{i1},\ldots,y_{in_i}|\mu,\theta,\tau^2,\sigma^2 \sim \text{i.i.d. Normal}(\theta_i,\sigma^2)$$

for $i = 1,\ldots,m$, and $j = 1,\ldots,n_i$.

   (a) Which do you think is bigger, $\mathbb{V}(y_{ij}|\theta_i,\sigma^2)$ or $\mathbb{V}(y_{ij}|\mu,\tau^2)$? Provide an intuitive answer without doing any calculation.

   (b) Do you think $\mathbb{C}(y_{ij},y_{ij'}|\theta_i,\sigma^2)$ is positive, negative, or zero? Answer the same for $\mathbb{C}(y_{ij},y_{ij'}|\mu,\tau^2)$.

   (c) Now compute each of the following quantities and compare to your answers in parts (a) and (b).

      i. $\mathbb{V}(y_{ij}|\theta_i,\sigma^2)$
      ii. $\mathbb{V}(\bar{y}_{ij}|\theta_i,\sigma^2)$
      iii. $\mathbb{C}(y_{ij},y_{ij'}|\theta_i,\sigma^2)$ $(j \neq j')$
      iv. $\mathbb{V}(y_{ij}|\mu,\tau^2)$
      v. $\mathbb{V}(\bar{y}_{ij}|\mu,\tau^2)$
      vi. $\mathbb{C}(y_{ij},y_{ij'}|\mu,\tau^2)$ $(j \neq j')$

   (d) Show that if we have a prior $p(\mu)$ for μ, then

$$p(\mu|\theta_1,\ldots,\theta_m,\sigma^2,\tau^2,y) = p(\mu|\theta_1,\ldots,\theta_m,\tau^2).$$

   Provide a verbal interpretation of this result.

2. Use the model in question 1 with the following data where $m = 5$:

i	n_i	y_{i1}	y_{i2}	y_{i3}	y_{i4}	y_{i5}	y_{i6}
1	2	1.75	-0.52				
2	4	-1.77	-1.46	-3.21	0.51		
3	4	1.97	-2.35	-0.85	0.06		
4	4	1.11	-0.80	1.34	0.64		
5	6	1.87	1.44	1.38	2.45	0.73	1.44

   (a) Estimate θ_i using the sample mean $\bar{y}_{i\cdot} = n_i^{-1}\sum_{j=1}^{n_i} y_{ij}$.

   (b) Using a Uniform$(-6,6)$ prior for μ and independent Scaled Inverse Chi-squared$(1,1)$ priors for σ^2 and τ^2, estimate the posterior means for the parameters using JAGS. Call the posterior means $\hat{\theta}_i$ for $i = 1,\ldots,5$.

   (c) These data were simulated using $\theta = (0.36,-0.73,0.04,0.11,1.43)$. Compare the sample means $\bar{y}_{i\cdot}$ and the posterior means $\hat{\theta}_i$ to θ_i. (Use a scatterplot.)

   (d) Draw 95% posterior credible intervals onto the scatterplot in part (c).

Note: JAGS (unlike Stan) does not accept ragged arrays. I would suggest that you make `y = c(1.75, -0.52, -1.77,...)` into a vector of 20 elements. You can then create a variable `group <-rep(1:5, c(2,4,4,4,6))` so that `y[i] ~dnorm(theta[group[i]], sigma2inv)` for $i = 1, \ldots, 20$.